

Дерево вывода:

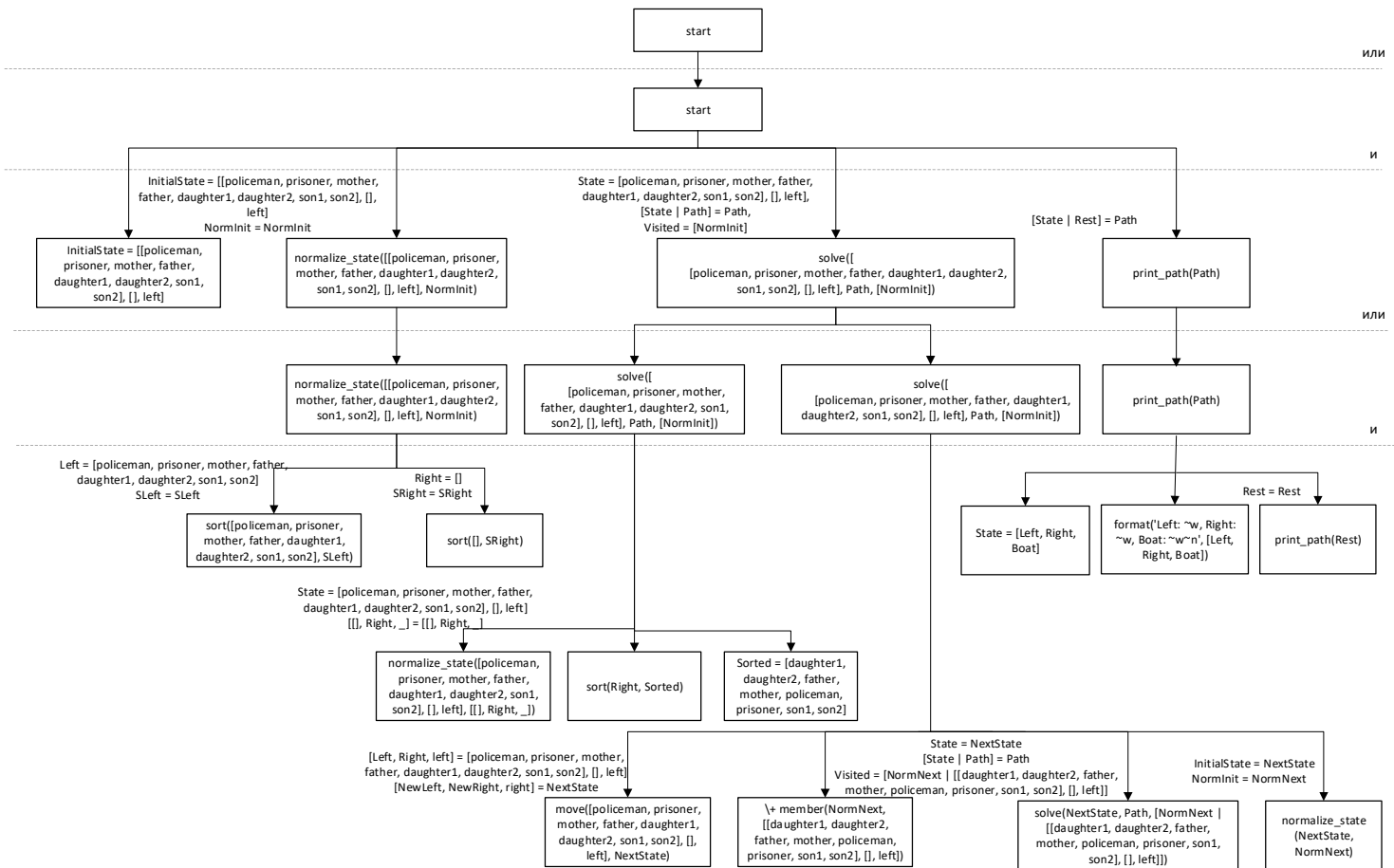


Рис. 1 - Дерево вывода

Тестирование:

```
?- start.
Left: [policeman,prisoner,mother,father,daughter1,daughter2,son1,son2], Right: [], Boat: left
Left: [mother,father,daughter1,daughter2,son1,son2], Right: [policeman,prisoner], Boat: right
Left: [policeman,mother,father,daughter1,daughter2,son1,son2], Right: [prisoner], Boat: left
Left: [mother,father,daughter2,son1,son2], Right: [policeman,daughter1,prisoner], Boat: right
Left: [policeman,prisoner,mother,father,daughter2,son1,son2], Right: [daughter1], Boat: left
Left: [policeman,prisoner,father,son1,son2], Right: [mother,daughter2,daughter1], Boat: right
Left: [mother,policeman,prisoner,father,son1,son2], Right: [daughter2,daughter1], Boat: left
Left: [policeman,prisoner,son1,son2], Right: [mother,father,daughter2,daughter1], Boat: right
Left: [father,policeman,prisoner,son1,son2], Right: [mother,daughter2,daughter1], Boat: left
Left: [father,son1,son2], Right: [policeman,prisoner,mother,daughter2,daughter1], Boat: right
Left: [mother,father,son1,son2], Right: [policeman,prisoner,daughter2,daughter1], Boat: left
Left: [father,policeman,prisoner,son1,son2], Right: [mother,daughter2,daughter1], Boat: right
Left: [father,son1,son2], Right: [mother,policeman,prisoner,daughter2,daughter1], Boat: left
Left: [son2], Right: [father,son1,mother,policeman,prisoner,daughter2,daughter1], Boat: right
Left: [policeman,prisoner,son2], Right: [father,son1,mother,daughter2,daughter1], Boat: left
Left: [prisoner], Right: [policeman,son2,father,son1,mother,daughter2,daughter1], Boat: right
Left: [policeman,prisoner], Right: [son2,father,son1,mother,daughter2,daughter1], Boat: left
Left: [], Right: [policeman,prisoner,son2,father,son1,mother,daughter2,daughter1], Boat: right
true
```

Рис. 2 – Результат работы программы

Вывод:

В ходе выполнения лабораторной работы были приобретены навыки логического программирования поиска решения задачи; были изучены основы языка программирования “Prolog”. С помощью среды программирования “SWI - Prolog” была разработана программа для решения логической задачи.